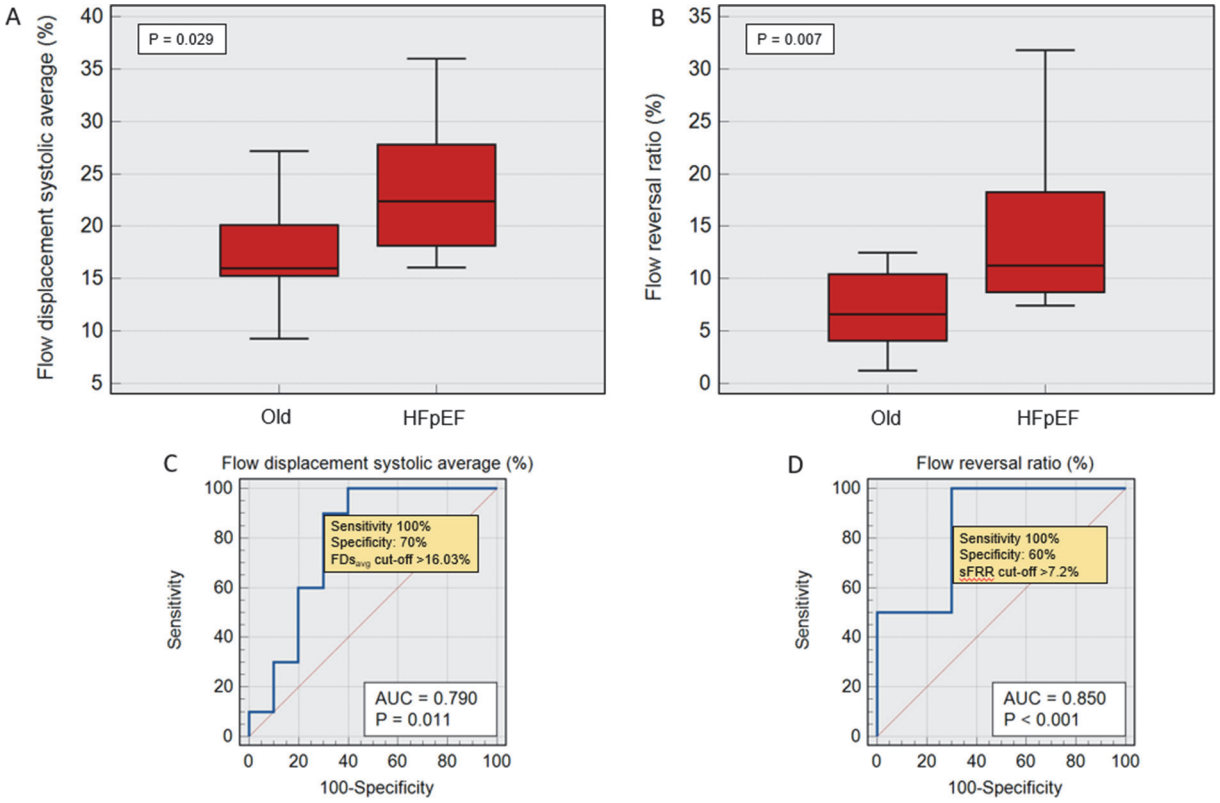

Abstract 31 Figure 3 **A** – Bar charts demonstrating average systolic flow displacement trends in age-gender-matched old healthy cohorts and HFpEF patients. **B** – Bar charts demonstrating systolic flow reversal ratio trends in age-gender-matched old healthy cohorts vs HFpEF patients. **C** and **D** – Receiver operating characteristic (ROC) with area under the curve demonstrating acceptable correlation in age-gender-matched old HCs vs patients with HFpEF

Conclusion Aortic flow haemodynamics (FDs_{avg} and sFRR) are significantly affected in ageing and HFpEF patients. Studies with larger and diverse cohort are required to draw definitive conclusions.

Acknowledgements

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32 SEX-SPECIFIC CARDIAC MAGNETIC RESONANCE PULMONARY CAPILLARY WEDGE PRESSURE MODEL PREDICTS OUTCOMES IN HEART FAILURE: A MULTI-CENTRE STUDY

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Introduction Non-invasive imaging is routinely used to estimate left ventricular (LV) filling pressure (LVFP) in heart failure (HF). Cardiovascular magnetic resonance (CMR) is emerging as an important imaging tool for sub-phenotyping HF and estimating pulmonary capillary wedge pressure (PCWP), a surrogate for LVFP.[1]

This research sought to investigate the following questions:

1) Does sex (male/female) influence CMR-derived PCWP?

2) Compared to a previously validated generic CMR-derived PCWP model,[1] does the use of a sex-specific model improve precision?

3) Is a novel sex-specific CMR-modelled PCWP prognostically significant?

Materials and Methods A derivation cohort (Norwich) involving patients with suspected HF underwent invasive right heart catheterisation and CMR within 24 hours of each other (*figure 1A*). This cohort was used to develop a sex-specific CMR model to estimate PCWP. The sex-specific CMR-derived PCWP model was developed using stepwise multivariate regression, using CMR variables significantly associated with PCWP during multivariate linear regression.

A validation cohort (Sheffield) of patients with confirmed heart failure underwent CMR (*figure 1B*) and was used to evaluate for the primary end-points of heart failure hospitalisation and MACE (composite cardiovascular death, HF hospitalisation, non- fatal stroke and myocardial infarction). A comparison in the prognostic performance between the generic model and the novel sex-specific CMR-derived PCWP models was performed.

Results 835 subjects were recruited from the Sheffield centre (derivation), and 454 subjects were recruited from the Leeds centre (validation).

Sex (male/female) does influence CMR-derived PCWP, since left atrial volume (LAV) and left ventricular mass (LVM) are higher in men (*figure 1C*). A novel sex- specific model incorporating LAV and LVM was created:

Sex-specific PCWP = 5.8 + (0.075*LAV) + (0.05*LVM) – (1.99*sex) [F=0; M=1]

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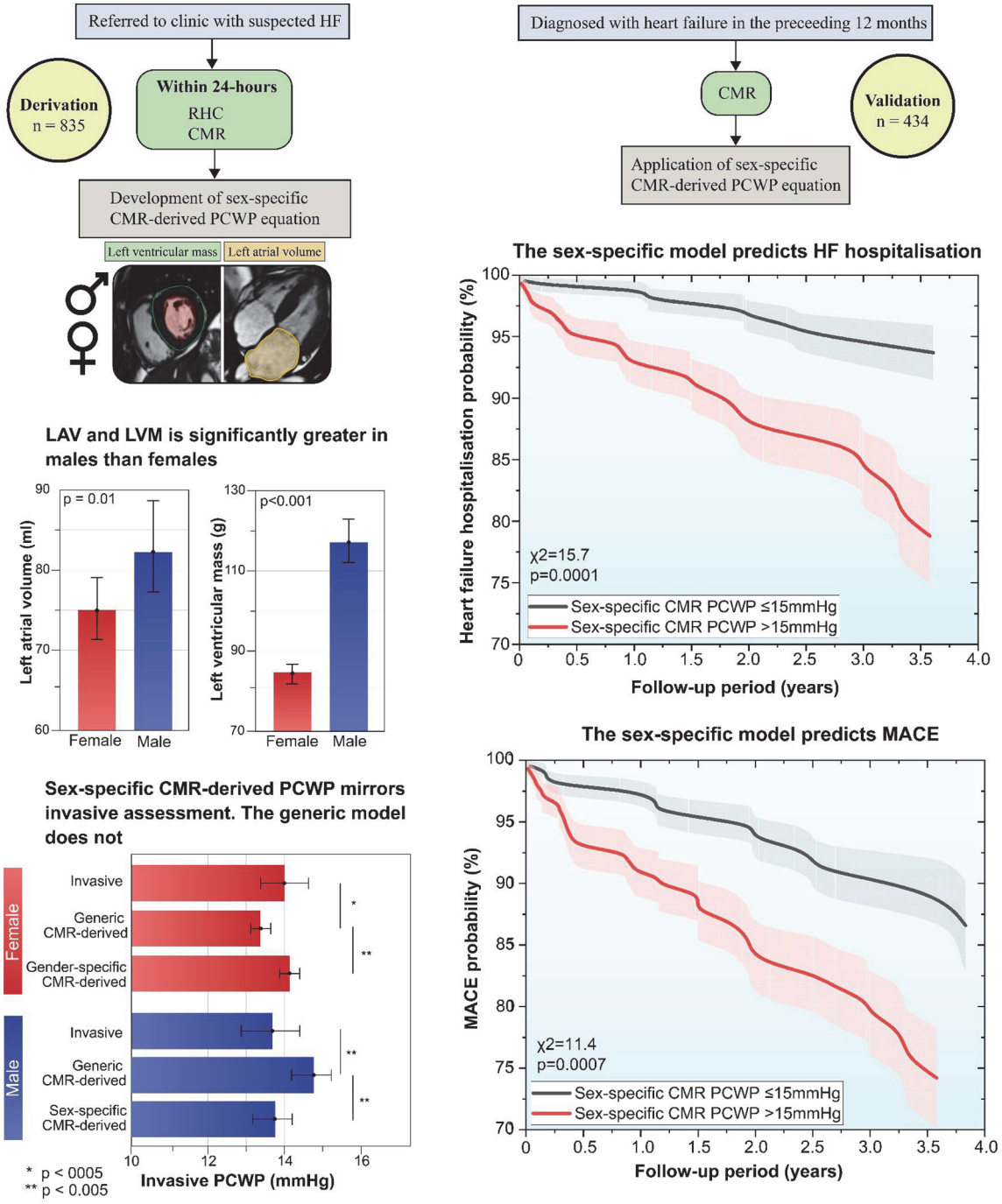

Abstract 32 Figure 1

Applying the generic CMR model resulted in significant differences in PCWP between males and females (14.7±4 versus 13±3 mmHg, p>0.001). Using the sex- specific model, CMR-derived PCWP was comparable between males and females (14.1±3 versus 13.8 mmHg, p=0.3), highlighting its excellent performance (*figure 1D*). Notably, there was no difference in invasive PCWP assessment between males and females (13.7 ±6 versus 14.0±6 mmHg).

In the validation cohort, the sex-specific model was associated with HF hospitalisation (hazard ratio 3.9, p=0.0002) and MACE (hazard ratio 2.5, p=0.001) over a mean follow-up period of 2.4±1.2 years (*figure 1E and F*). This was not the case for the generic CMR-derived PCWP.

Conclusion A sex-specific CMR-derived PCWP model improves precision and prognostication in heart failure compared to a generic CMR-derived PCWP model.

Acknowledgements Nil.

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